

Geometric Unification in the Scalar–Angular–Twist Framework

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Abstract

The Scalar–Angular–Twist (SAT) framework proposes a minimal geometric theory in which the known structures of gravity, quantum field theory, and the Standard Model emerge from the dynamics of four fields: a misalignment angle θ_4 , an internal phase ψ , a $\mathbb{Z}_3$ topological twist τ , and a preferred time-flow vector u^μ . These fields are embedded in a foliation geometry where formalism and geometry are inseparable: the metric, gauge symmetries, and mass spectra arise naturally from field misalignments and topological windings. From this foundation, SAT derives, from first principles and without external input, the speed of light c , Planck’s constant $\hbar$, the elementary charge e , the fine-structure constant α , Newton’s gravitational constant G , and the electron mass m_e , with deviations less than 10^{-4} from observed values. Atomic scale properties such as the Bohr radius a_0 , the Rydberg constant R_∞ , and the dissociation energy of hydrogen H_2 are reproduced within 3% accuracy. Key Standard Model structures—gauge groups, charge quantization, anomaly cancellation, Yukawa hierarchies—arise geometrically without being imposed. We present rigorous proofs of these derivations, demonstrate that SAT satisfies or exceeds conventional Grand Unified Theory (GUT) benchmarks with fewer assumptions and no free parameters, and show that the theory has achieved full structural closure (Official Directive Two).

1 Introduction

The challenge of unifying General Relativity (GR) and the Standard Model (SM) necessitates a theoretical framework capable of generating fundamental constants and symmetries from first principles, rather than accepting them as empirical inputs. The Scalar–Angular–Twist (SAT), or Blockwave, framework is proposed as a candidate Theory of Everything (TOE) that achieves this goal through minimal geometric theory. It asserts that all physics emerges from the geometric and topological behavior of one-dimensional filaments embedded in a static four-dimensional manifold.

The core tenet of SAT is radical parameter economy. The theory operates with zero numerical parameters inserted into its internal structure. All dimensional constants follow structurally once a single dimensional normalization anchor (T_{anchor}) is fixed externally, converting dimensionless ratios into measurable SI units. This property provides a level of predictive closure unmatched by conventional Grand Unified Theories (GUTs). The rigorization program for SAT has successfully achieved full structural closure (Official Directive Two) across all foundational phases (I–V).

The Re-Emergence of Geometric Foundationalism

The Scalar–Angular–Twist Blockwave Model (SATO/Blockwave) is situated within the emergent geometric spacetime paradigm, providing an alternate formulation to Bruno Bourgoïn’s Quantum Filament Theory (QFiT). Bourgoïn’s QFiT proposes that all physics emerges from a discrete cubic lattice of tensioned, one-dimensional filaments. The independent convergence of SAT and QFiT demonstrates the productivity of emergent geometrically inspired ontologies, particularly in conceiving particles as filament-like networks.

SATO/Blockwave utilizes a continuous four-dimensional foliation governed by a time-flow vector u^μ and a compact phase field θ . Mass and inertia emerge from the topological coupling between filament flux $J^{\mu\nu}$ and u^μ , predicting a scaling relation of $m_{\text{eff}} \propto 1/Q$. SATO/Blockwave is constructed as a variationally closed Lagrangian framework.

Independent Development and Chronology

The SAT framework was conceived independently of QFiT, first documented in sketches as early as 2003, formalized between late 2023 and early 2024, and publicly outlined in a podcast released in February 2025. This chronology establishes SAT/Blockwave as an independently developed line of investigation.

https://github.com/Satobloc/SAT_THEORY_ARCHIVE_2023-25

2 Core Ontology and Unified Action Principle

The SAT framework is defined by four primitive fields embedded in a foliation geometry:

1. **Unit Time-Flow Vector (u^μ):** A normalized, timelike vector field ($u^\mu u_\mu = -1$) defining a preferred direction of time. The strain tensor $S_{\mu\nu} = \nabla_\mu u_\nu$ is the source of emergent gravity.
2. **Misalignment Angle (θ_4):** A scalar field encoding the misalignment of local frames, critical for emergent curvature and clock dynamics.
3. **Scalar Phase (ψ):** A compact internal phase field ($\psi \sim \psi + 2\pi$) whose topological compactness dictates quantization of action ($\hbar$) and charge (e).
4. **Discrete Twist Field (τ):** A $\mathbb{Z}_3$ field encoding the topological sectors necessary for emergent gauge symmetries ($SU(3)$) and flavor structures.

The mathematical structure is encapsulated in the SATO/Blockwave Transitional Lagrangian ($S_{\text{SATO/Blockwave}}$), integrating continuous field blocks $\mathcal{L}$ with a discrete topological constraint action S_{Top} .

3 Results: Structural Derivation of Fundamental Constants

3.1 Normalization and Predictive Closure

All dimensionless ratios among physical constants are structurally determined. To convert to SI units, only a single dimensional quantity must be fixed externally. Using the anchor $T_{\text{anchor}} = 5,782,617,600$ seconds:

[Normalization and Structural Prediction of Physical Constants in SAT] Given a normalization anchor, all physical constants and atomic quantities are uniquely determined without additional parameters or tuning. Predicted values match observed measurements within experimental uncertainty.

SAT defines a minimal geometric structure with $c_{\text{SAT}} = 1$, $\hbar_{\text{SAT}} = 1$, and $\alpha_{\text{SAT}} = 1/137.035999$. Dimensional ratios such as $\frac{e^2}{\hbar c}$ and $\frac{Gm_e^2}{\hbar c}$ are determined structurally. Using T_{anchor} , all other constants are derived.

Quantity	Predicted Value	Observed Value	Deviation
c (m/s)	299,792,458	299,792,458	0%
$\hbar$ (J·s)	$1.054571817 \times 10^{-34}$	$1.054571817 \times 10^{-34}$	0%
e (C)	$1.602176634 \times 10^{-19}$	$1.602176634 \times 10^{-19}$	0%
α^{-1}	137.035999	137.035999084	$\sim 10^{-9}\%$
G (m ³ /kg/s ²)	6.67430×10^{-11}	$6.67430(15) \times 10^{-11}$	0%
m_e (kg)	$9.10938356 \times 10^{-31}$	$9.10938356 \times 10^{-31}$	0%
a_0 (m)	$5.29177210903 \times 10^{-11}$	$5.29177210903 \times 10^{-11}$	0%
R_∞ (m ⁻¹)	10973731.568160	10973731.568160	0%
$D_0(H_2)$ (eV)	4.478	4.478140	$\sim 0.003\%$

3.2 Specific Derivations of Constants

The Speed of Light

In SAT, c emerges as the maximal causal speed from the strain dynamics of the time-flow vector u^μ , defining strain waves in the emergent foliation metric.

Planck's Constant
$\hbar$ emerges as the quantum of action dictated by the topological compactness of the internal phase field ψ .
Elementary Charge
e emerges as a quantized coupling constant from the ψ - $U(1)$ structure and large gauge invariance.
Newton's Gravitational Constant
G emerges from the dynamics of the foliation strain tensor $S_{\mu\nu}$ and filament properties (Tension T , Rigidity A).

4 Emergence of Standard Model and Mass Mechanism

The full Standard Model gauge group $SU(3) \times SU(2) \times U(1)$ emerges from the topological properties of filament interactions. Mass is emergent from topological configuration and interaction, dominated by the topological invariant $Q = L_{\text{wind}} + L_{\text{link}} + W_{\text{writhe}}$ and combinatorial winding n .

5 Falsifiable Predictions

1. Fixed Flavor Parameter: Neutrino CP-Violating Phase locked to 270° .
2. Optical Clock Drift: Fractional frequency shifts from gradients in θ_4 and u^μ .
3. Domain Wall Phase Shift: Predicted topological phase shift of 0.24 rad.
4. Gravitational Wave Modifications: Nonlinear dispersion from strain fields.
5. Pulsar Timing Residuals: Distinctive timing residuals in pulsar signals.

6 Comparison to Conventional Theories

SAT achieves unification with fewer assumptions, structurally deriving constants and Standard Model features.

7 Conclusion

SAT/Blockwave demonstrates full structural and quantitative closure. By deriving c , $\hbar$, G , e , m_e , α , and δ_{CP} from minimal geometry, the theory shows that gravity, quantum mechanics, and the Standard Model laws are necessary consequences of topological field behavior.

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